

Female Reservation and the Qualifications of Local Elected Officials in India

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Working draft

Abstract

How does reserving an office for women change the qualifications of the person elected? We examine rural panchayats in Bihar, Uttar Pradesh, Rajasthan and Kerala, and urban councillors in Mumbai and Delhi. Education is measured consistently by the share with a university degree. Among rural village heads, the adjusted reserved-minus-open difference is -15.0 percentage points (95% CI -16.5 to -13.4) in Bihar, and negative in each UP election. Rajasthan's classified-winner estimate is negative, but missing education prevents an all-winner conclusion. Kerala's village ward members show smaller, imprecise differences. In Mumbai, reserved-seat councillors have a higher estimated graduate share, with an interval that includes zero, and fewer pending criminal cases. Delhi's graduate-share intervals include zero in three elections, while pending-case differences are negative. These comparisons control for geography and, except in Mumbai, caste reservation. Their causal interpretation requires conditional comparability of reserved and open seats. The evidence concerns qualifications at election; it does not establish how reservation changes performance in office.

1 Introduction

Reserving elected offices for women changes who can contest them. Does it also change the qualifications of the people elected? The relevant comparison is between winners of women-reserved and open seats. Open seats can elect women, so a comparison of female and male politicians does not answer this question.

Existing rural studies often find lower schooling and less political experience under reservation. Their evidence spans different offices, outcomes and research designs. Some compare candidates rather than winners; others compare women in reserved seats with men in open seats. These distinctions determine which findings bear on the reservation effect on elected elites.

This paper brings that evidence together with administrative records of elected officials. The rural analyses cover village heads and council members in Bihar, Uttar Pradesh, Rajasthan and Kerala; the urban analysis covers Mumbai and Delhi councillors. Graduate share provides a common education measure. The comparisons remain separate by office and election, and rural panchayats remain separate from urban councils.

2 Why these attributes matter

Local officials help prepare budgets, pursue public works and allocate benefits. These tasks require reading documents, understanding rules and dealing with other officials (Auerbach et al. 2025,

708–10; Joshi et al. 2025, 56). Literacy can make those tasks easier, while further schooling may supply additional skills. But literacy, secondary schooling and a university degree measure different levels of education; the evidence does not establish that each step yields the same improvement in governing ability.

In South India, more educated politicians have a smaller advantage in obtaining subsidized-food eligibility cards for their own households, and landless households are better targeted under more educated council heads. These are observational associations with public-resource allocation (Besley et al. 2007, Tables 4–5). In Uttarakhand, secondary-educated and graduate heads report higher panchayat performance scores than illiterate heads. That study uses self-assessments and education-category comparisons without other controls; it supplies suggestive evidence, not a causal estimate of schooling (Joshi et al. 2025, 57–58, Table 3).

Evidence on degrees is also mixed. A close-election study of Indian state legislators finds higher night-time-light growth when a graduate narrowly defeats a non-graduate (Jain et al. 2022, Table 2). Another finds improved education outcomes under graduate legislators only in more developed states, with no clear overall benefit (Lahoti and Sahoo 2020). Across national leaders, US legislators and Brazilian mayors, college graduates show no consistent performance advantage (Carnes and Lupu 2016). These findings motivate measuring educational attainment, but do not make graduate status a sufficient measure of competence or establish its value in every local office.

Political experience may help officials learn how government works. In Rajasthan’s urban councils, education and prior election victories predict greater procedural knowledge. Knowledge, in turn, is associated with greater reported constituent engagement and access to higher officials; the main budget-involvement estimate is imprecise. Much of the activity evidence is self-reported, and these associations do not isolate a causal effect of education or experience. A close-election comparison finds only modest, often imprecise knowledge gains from winning office over the following eighteen months (Auerbach et al. 2025, Table 3 and Figures 5–6).

Pending criminal cases can signal risks to public integrity. In Indian state assemblies, narrowly electing an accused candidate is followed by lower economic growth measured through night-time lights and weaker road provision, with larger losses for serious or financial accusations (Prakash et al. 2019). This evidence makes criminal records relevant to evaluating entrants, but it concerns state legislators. A pending charge remains an allegation; its seriousness and the process generating charges matter. Our Mumbai and Delhi indicators measure any pending case and cannot distinguish these dimensions.

Age can reflect experience and the length of a politician’s remaining career. Evidence from Italian municipalities shows that younger mayors facing reelection engage more in electoral budget cycles, without clear advantages in house prices or the speed of public-good provision (Alesina et al. 2019, Tables 5–7). Age therefore describes a potentially consequential change in elite composition, without supplying a general ranking of competence.

Income and wealth capture economic resources and social position. Higher household income is associated with greater procedural knowledge in Rajasthan’s urban-politician survey (Auerbach et al. 2025, Table 3), but that association does not show that richer officials govern better. Wealth observed after election can also reflect returns to office: Indian state-election winners accumulate assets faster than runners-up, including in close elections (Fisman et al. 2014). Income is a flow, assets a stock, and neither is equivalent to honesty or ability. This draft estimates reservation effects on education, age and pending cases in Mumbai and Delhi; income and wealth remain potential extensions.

3 Existing evidence

Lower schooling and less experience under reservation appear in several rural studies. In the 2001 working paper’s West Bengal sample, reserved council heads have 7.13 rather than 9.92 years of schooling, a difference of -2.79 years (SE 0.54); 11% rather than 43% had previously been elected to the council (Chattopadhyay and Duflo 2001, Table 3). In the subsequent Birbhum council term, the adjusted schooling gap is -2.10 years (SE 0.55), and reserved leaders are 26.5 percentage points more likely to be first-time pradhans (SE 6.8) (Beaman et al. 2009, Table II, column 6). These samples overlap geographically and are not independent national tests.

The May 2007 southern-state draft, published in 2008, reports -2.62 years of schooling (SE 0.77) and a political-knowledge gap of -1.41 points (SE 0.29). That regression excludes caste-reserved councils (Ban and Rao 2007, Table 9). Karnataka’s descriptive comparison reports degrees among none of the 23 reserved female presidents and among 19% of the 48 male presidents. Illiteracy is 9% and 4%, respectively; the comparison also conditions on the leader’s gender, so it is not a clean reservation contrast (Raabe et al. 2009, Tables 3–4).

The 2011 working paper on the NCAER village panel, published in 2015, reports illiteracy of 24.72% in reserved observations versus 5.74% in unreserved observations, secondary education or above of 28.65% versus 62.69%, and previous officeholding of 11.67% versus 24.24%. These are descriptive comparisons across 180 reserved and 459 unreserved observations (Deininger et al. 2011, Table 3). The 2013 Andhra Pradesh working paper, published in 2017, reports that 15% of reserved heads versus 54% of unreserved heads have higher-secondary schooling or more; 11% versus 22% have prior political experience. The characteristics table covers 128 reserved and 169 unreserved heads (Afridi et al. 2013, Table 2).

Recent studies require additional distinctions. The September 2022 Karnataka candidate draft reports lower absolute qualification distributions under reservation (Mori et al. 2022, fig. 2 and Table 3). The published article reports stronger cognitive selection relative to the eligible female population (Mori et al. 2025). That relative benchmark does not establish that reservation increases the elected elite’s absolute qualifications. It also concerns candidates. The March 2026 Bihar threshold draft reports standardized schooling effects of -0.765 SD for candidates (SE 0.024; $N=44,583$) and -0.921 SD for winners (SE 0.073; $N=3,294$). Substituting husbands’ schooling gives -0.147 SD and -0.364 SD, respectively (Bamezai et al. 2026, Table 7). Spousal schooling does not reveal who governs.

Favorable judgments about participation, voter attitudes or policy performance address different outcomes. They can coexist with lower schooling among entrants. Likewise, a positive selection score relative to same-gender citizens can coexist with lower absolute schooling. A synthesis of the reservation effect must keep the outcome, elected status, comparison group and office fixed before combining estimates. The studies reviewed here do not yet supply a harmonized meta-analysis.

4 Data and empirical strategy

4.1 Outcomes and samples

The target is the impact of female reservation on the elected officeholder’s qualification, relative to opening the same office to either sex. An open seat can elect a woman. Comparing female and male winners would answer a different question, as would comparing all candidates rather than the people elected.

The main education outcome is an indicator for a completed university degree or above. Its mean is the graduate share. Illiteracy describes the lower end of the same education distribution and appears alongside graduate share in the appendix. These indicators are mutually exclusive but not exhaustive: a literate non-graduate belongs to neither. Their estimates must not be counted as independent measures of quality or independent studies. Unknown qualifications remain missing.

Separate linear regressions for these binary outcomes estimate changes in their shares; they do not assume that the outcomes are statistically independent. On an identical sample with identical controls, coefficients for a complete set of education categories would sum to zero. Testing a change in the entire education distribution would require a joint test that accounts for the covariance across category estimates. The paper reports pointwise intervals for individual shares, without making that joint claim.

4.2 Specification and inference

For each rural state, election and office, the analysis estimates

$$Y_i = \alpha + \beta R_i + \gamma_{g(i)} + \delta_{c(i)} + \epsilon_i,$$

where R_i records female seat reservation, $\gamma_{g(i)}$ controls for geography and $\delta_{c(i)}$ for caste reservation. Geography is the block when it is populated throughout the relevant slice, and otherwise the district. The district and block together identify a block. Standard errors are clustered at the geographic-control level. All reported intervals are pointwise 95% intervals from `fixest`, with its small-sample variance adjustment and a Student t critical value based on the number of clusters minus one. They do not adjust for testing several offices or outcomes. Kerala’s village and district-member regressions have only 14 district clusters. Their intervals use a few-cluster approximation and warrant particular caution.

For the binary education outcome, 100β is the adjusted reserved-minus-open difference in percentage points. The linear model describes this difference without imposing equal distances between school qualifications. Its causal interpretation requires conditional comparability of reservation assignments and adequate outcome coverage. Geographic and caste controls alone do not verify that requirement. No lottery or discontinuity design is claimed for these administrative-data regressions. Mumbai uses administrative-ward and council effects, with standard errors clustered by electoral ward. That specification does not include caste-reservation controls. Delhi uses separate election-year regressions with assembly-constituency and caste-reservation effects, clustering by assembly constituency. It therefore follows the rural specification’s geographic adjustment and inference convention, using urban electoral geography. Ward boundaries change across elections; no Delhi ward panel is assumed.

The administrative inputs flow from state election repositories through `local_elections`. The analysis records input hashes, winner selection, exclusions, outcome classification and regression samples. Bihar’s winners are those the state release derives from the results form: the highest recorded vote, an uncontested candidacy, or a tie drawn by lot. The release names no winner where the form cannot decide, such as a candidate listed twice with different votes or a tie without a lot mark; those seats are excluded. UP’s earlier files report elected *pradhans*; its latest candidate file marks winners. Rajasthan links nomination characteristics to results. Kerala records elected ward members, including those who subsequently hold council leadership positions. The reservation variable for those records refers to their ward, not the presidency.

5 Rural panchayats

5.1 Village heads

Rural village heads: graduate or above

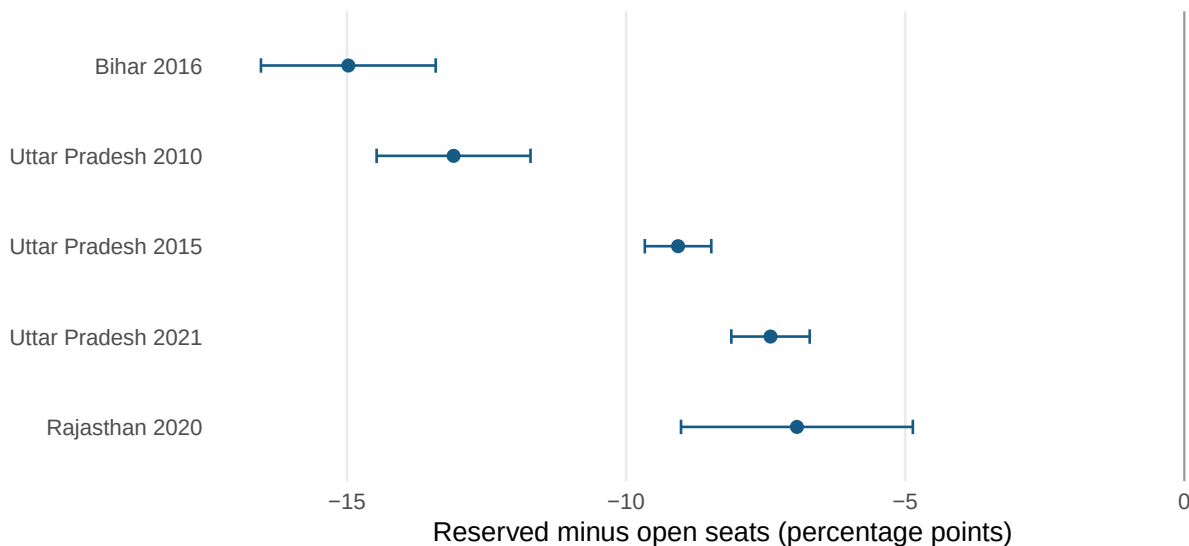


Figure 1: Graduate-share differences among rural village heads. Points compare women-reserved with open seats, conditional on geography and caste reservation; bars are pointwise 95% confidence intervals clustered by geography. UP 2010 uses districts; other rows use blocks. Education is restricted to classified qualifications. Ambiguous winners and unknown reservation are excluded. Sources: state election records through the central election repository.

UP’s graduate-share gap is -13.1 percentage points (95% CI -14.5 to -11.7) in 2010, -9.1 percentage points (95% CI -9.7 to -8.5) in 2015, and -7.4 percentage points (95% CI -8.1 to -6.7) in 2021. These are separate election samples. Changes in their coefficients do not by themselves identify learning, convergence or an effect of repeated reservation.

Rajasthan’s estimate is -6.9 percentage points (95% CI -9.0 to -4.9). Its source includes many qualifications labelled “Other”, which remain unclassified. The estimate therefore describes winners whose graduate status can be classified. Allowing every missing qualification to take either degree status puts the adjusted difference for all eligible winners between -29.7 and 31.5 percentage points. These data alone do not determine its direction.

5.2 Bihar’s other offices

Bihar provides evidence across village, block and district offices. Its analysis includes 210,717 elected seats; seats whose results form cannot identify a winner are excluded. Unknown education remains missing. Implausible or missing ages do not remove a winner from education regressions.

Bihar 2016: graduate or above

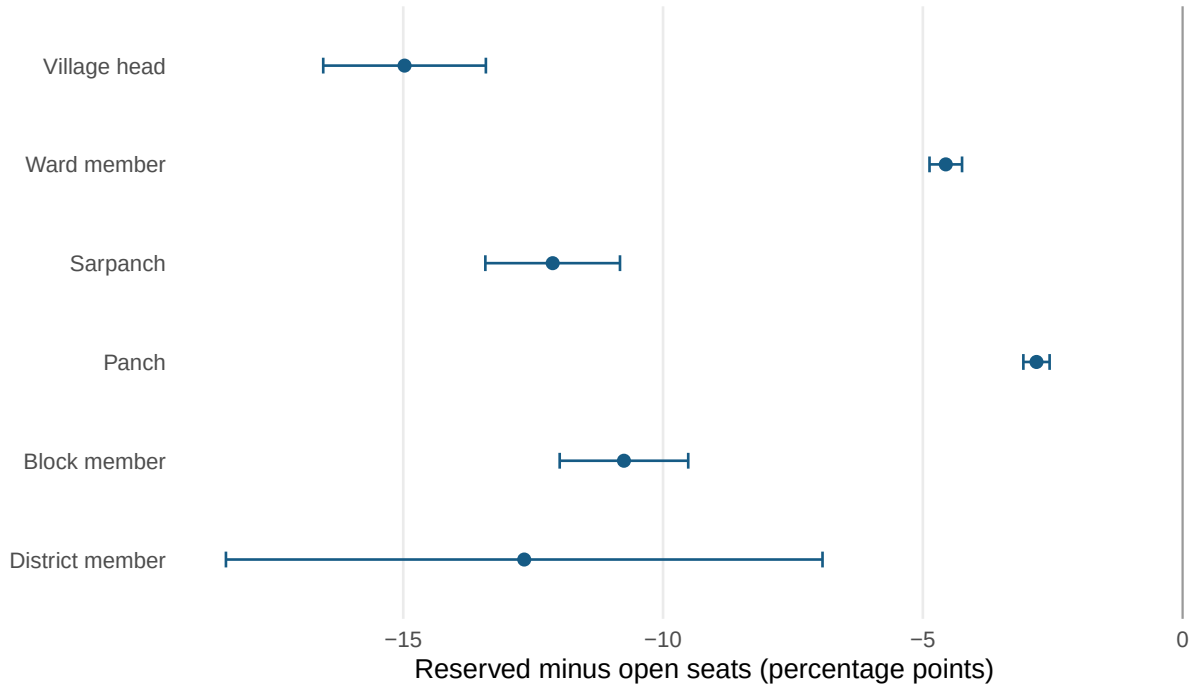


Figure 2: Bihar 2016: graduate-share differences by office. Points are reserved-minus-open differences with block and caste-reservation controls, except district council members, who use districts. Bars are pointwise 95% confidence intervals clustered at that geographic level. Unknown education and ambiguous winner records are excluded. Source: Bihar SEC candidate records through the central election repository.

The graduate-share estimate is negative in every office. The appendix reports illiteracy alongside graduate share as another part of the education distribution. Separate age regressions describe younger winners under reservation.

Bihar's 2021 election adds 244,474 elected seats across the same six offices. Its candidate records include age but not education, so 2021 contributes age regressions only. Reservation comes from the State Election Commission's feed for the 2021–2026 term, whose women-reserved village-head seats match the Commission's 2021 report. Seats with a single nominee and no result record count their nominee as the winner, as 2016 counts uncontested winners. Reserved-seat village heads are 3.2 years younger in 2021 (95% CI 2.7 to 3.6), against 3.6 years in 2016; the appendix reports every office.

5.3 Kerala by election and office

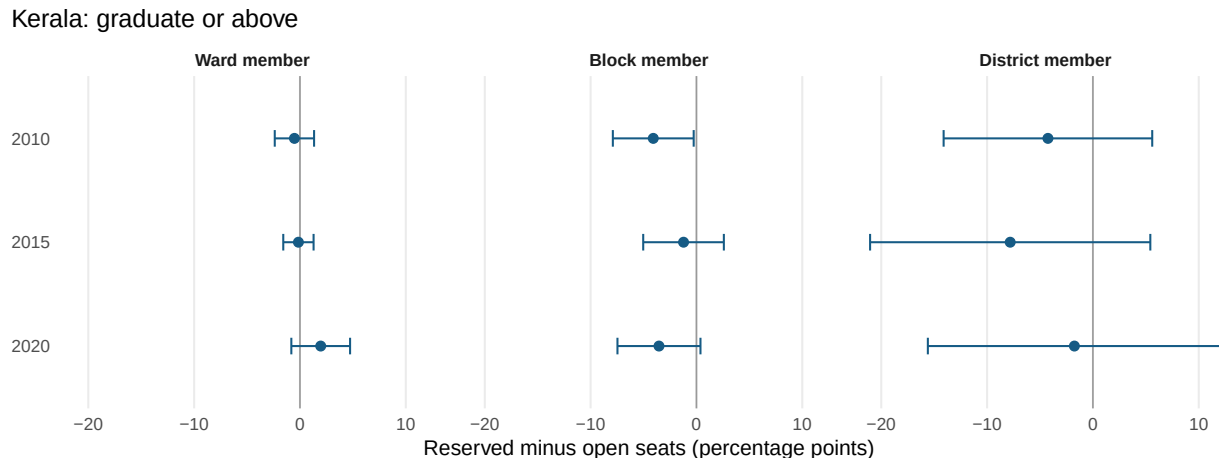


Figure 3: Rural Kerala: graduate-share differences by election and office. Each point comes from a separate regression with caste-reservation controls. Block council members use block effects and clustering; village and district ward members use districts because block identifiers are unavailable. Bars are pointwise 95% confidence intervals. Unclassified qualifications are missing. The common horizontal scale permits comparison across offices. Source: Kerala election records through the central election repository.

For Kerala’s village ward members, the differences are -0.5 percentage points (95% CI -2.4 to 1.3), -0.1 percentage points (95% CI -1.6 to 1.3), and 2.0 percentage points (95% CI -0.8 to 4.7). They provide no precise evidence of the large negative gaps observed among village heads elsewhere. They concern ward members, however, so the contrast with Bihar and UP cannot be attributed to the state alone.

Education is recorded as free text. The coding recognizes reported degrees and school qualifications, leaves incomplete or unclassified qualifications missing, and exports every source label with its classification and frequency. It does not infer that an unmatched string means no degree. These rules differ from the earlier pooled Kerala descriptive table, which used loose substring matches and included unmatched values in its denominator. That historical table is preserved in the repository’s evidence archive rather than treated as an estimate from this specification.

6 Urban councils

6.1 Mumbai

Mumbai’s data link reservation to councillor qualifications in the Praja/affidavit records held in the Karekurve-Ramachandra and Lee replication deposit (Karekurve-Ramachandra and Lee 2023). The sample contains one observation per councillor spell with age recorded. Repeated survey waves do not create additional councillors. The analysis pools the 2012 and 2017 councils with council and administrative-ward effects; standard errors are clustered by electoral ward.

Urban: Mumbai councillors, 2012 and 2017

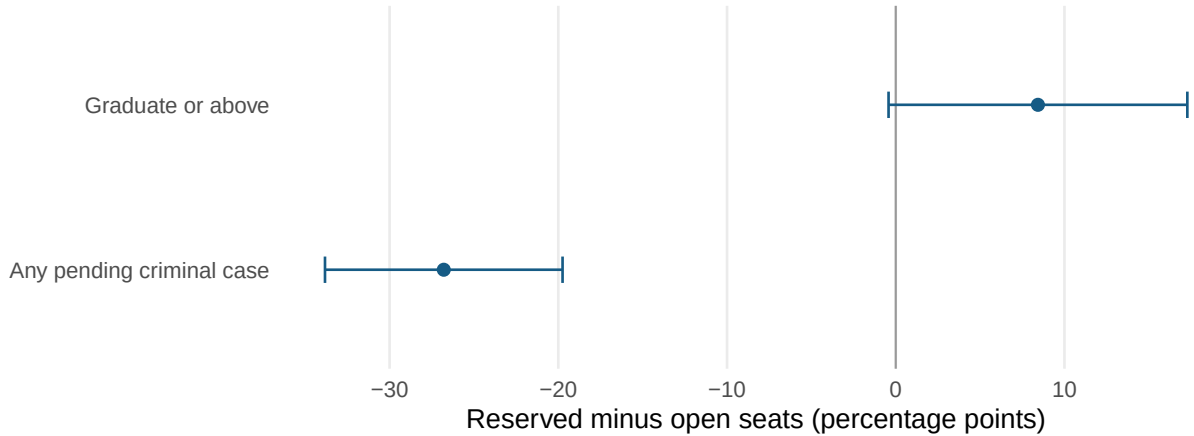


Figure 4: Urban Mumbai: qualifications of elected councillors in the 2012 and 2017 councils. Points are women-reserved-minus-open differences with administrative-ward and council effects; bars are pointwise 95% confidence intervals clustered by electoral ward. One observation is retained per councillor spell with age recorded. Pending cases are allegations, not convictions. Source: Praja/affidavit data in the Karekurve-Ramachandra and Lee replication deposit.

The graduate-share difference is 8.4 percentage points (95% CI -0.4 to 17.3). The pending-case difference is -26.8 percentage points (95% CI -33.8 to -19.7). The education interval permits a small negative effect as well as a positive one; it does not establish that reservation raises education. Pending criminal cases are separately reported because a charge does not establish guilt.

The earlier Rajasthan municipal comparison and Kerala’s urban descriptive tables belong to this urban evidence, not the rural estimates above. Rajasthan’s category “high school or less” includes the twelfth standard; Mumbai’s ends at the tenth. Those levels cannot be pooled as a common outcome. The historical tables remain available in the evidence archive, with their original denominators and qualifications.

6.2 Delhi

Delhi adds three urban elections with a common outcome definition and geographic adjustment. The winner roster and assembly mapping for 2012 and 2017 come from Goyal’s replication deposit (Goyal 2023). For 2022, the OpenCity/Lok Dhaba result compilation supplies winners, and the SEC gazette supplies reservation and assembly identifiers (Vaidyanathan 2023; State Election Commission, NCT of Delhi 2022, Annexure B). All 2022 reservation labels agree between the compilation and the gazette. The 2012 labels also match the official schedule (State Election Commission, NCT of Delhi 2012, Annexures I–III).

Qualifications are extracted directly from MyNeta’s election-specific winner profiles, with source URLs and cached responses retained (Association for Democratic Reforms 2026). Each linked profile identifies a winner in the same ward and agrees with the roster’s party. Missing profiles and unspecified schooling remain missing. The unlinked 2022 winners are the same two omitted by ADR because clear, complete affidavits were unavailable when its report was prepared (Association for Democratic Reforms and Delhi Election Watch 2022, 3). Pending cases are read from the accusation or pending-case tables, excluding the separate conviction tables. MyNeta’s headline criminal-case total can include both and is not used as the pending-case outcome.

Urban: Delhi councillors

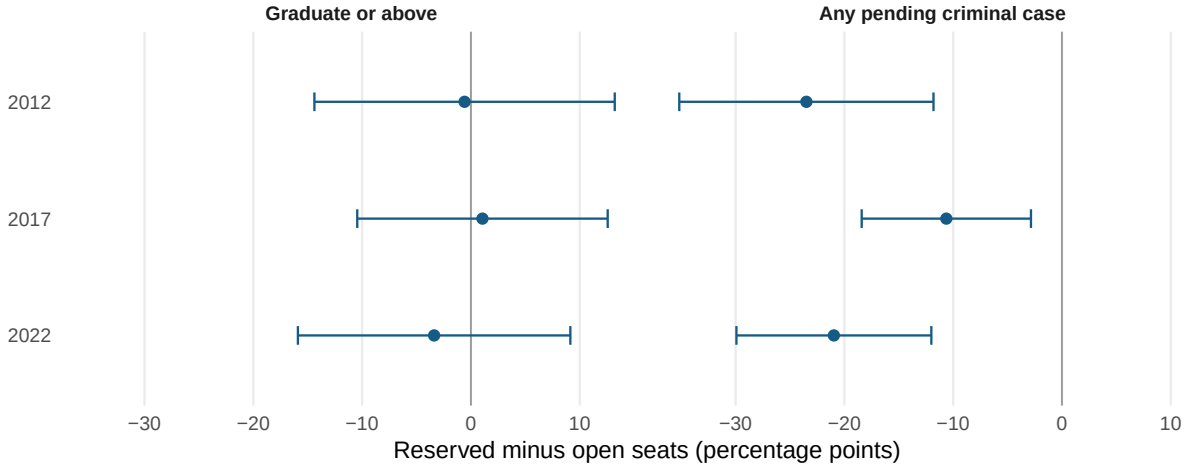


Figure 5: Urban Delhi: separate election-year estimates with assembly-constituency and caste-reservation effects. Bars are pointwise 95% confidence intervals clustered by assembly constituency. Each outcome uses winners with that attribute classified. Pending cases are allegations, not convictions. Sources: Goyal’s replication deposit, OpenCity/Lok Dhaba results, Delhi SEC reservation schedules and MyNeta winner profiles.

The graduate-share differences are -0.6 percentage points (95% CI -14.4 to 13.2), 1.1 percentage points (95% CI -10.4 to 12.6) and -3.4 percentage points (95% CI -15.9 to 9.1) in 2012, 2017 and 2022, respectively. None of the intervals establishes the direction of the education effect. Missing-data bounds permit either sign in the first two elections. The 2022 all-winner coefficient remains negative under every completion of missing education, but those bounds omit sampling uncertainty; they do not overturn the regression interval’s uncertainty about the sign.

Pending-case differences are -23.5 percentage points (95% CI -35.2 to -11.8), -10.6 percentage points (95% CI -18.4 to -2.8) and -21.0 percentage points (95% CI -29.9 to -12.0), respectively. All three remain negative under every binary completion of missing case records. These results concern recorded allegations, not proven misconduct or overall governing ability. The appendix reports age, education thresholds, coverage and missing-record bounds.

Source reconciliation matters here. The older local CSVs agree with Karekurve-Ramachandra and Lee’s deposited graduate coding (local_reservations contributors 2022a, 2022b; Karekurve-Ramachandra and Lee 2020). Against the currently retrieved profiles, however, graduate status differs for 86 of 185 jointly classified winners in 2012 and 1 of 262 in 2017. The analysis therefore uses the source profiles consistently across elections. This audit establishes disagreements with the older coding, without identifying the origin of those differences. The profiles themselves summarize self-declared affidavits; the underlying scans have not been independently re-read.

7 Discussion

The clearest negative education differences occur among rural village heads in Bihar and UP. Rajasthan’s missing qualifications prevent an all-winner conclusion. Kerala supplies evidence on different offices, with smaller ward-member estimates and wider uncertainty about their signs. Mumbai supplies an urban comparison in which the graduate-share estimate is positive but imprecise, while

the pending-case estimate is negative. Delhi supplies three further urban elections with imprecise graduate-share differences and negative pending-case estimates. The paper does not estimate an urban-rural difference or a national average.

A quantitative synthesis would need a common outcome and office, comparable reservation contrasts, and an accounting of shared samples. Schooling years, graduate shares, political knowledge and same-gender selection scores cannot be averaged as though they measured the same quantity. Multiple education thresholds from one sample also do not create independent evidence. The current evidence supports an organized comparison; a pooled reservation-effect estimate requires further harmonization.

8 Appendix: measurement and reproducibility

Table 1: Education coverage among eligible winner records. N includes unclassified qualifications.

State	Year	Office	Seat	Classified (%)	N
UP	2010	Village head	Open	99.6	34,654
UP	2010	Village head	Women	99.4	17,166
UP	2015	Village head	Open	100.0	39,093
UP	2015	Village head	Women	100.0	19,926
UP	2021	Village head	Open	95.3	32,997
UP	2021	Village head	Women	94.1	16,774
Rajasthan	2020	Village head	Open	81.8	5,695
Rajasthan	2020	Village head	Women	56.9	5,433
Kerala	2010	Block member	Open	91.0	983
Kerala	2010	Block member	Women	93.0	1,095
Kerala	2010	Ward member	Open	89.7	8,047
Kerala	2010	Ward member	Women	90.8	8,594
Kerala	2010	District member	Open	87.7	162
Kerala	2010	District member	Women	94.7	169
Kerala	2015	Block member	Open	87.4	1,021
Kerala	2015	Block member	Women	88.7	1,041
Kerala	2015	Ward member	Open	87.6	7,725
Kerala	2015	Ward member	Women	89.7	8,196
Kerala	2015	District member	Open	86.0	164
Kerala	2015	District member	Women	92.2	167
Kerala	2020	Block member	Open	88.7	980
Kerala	2020	Block member	Women	89.3	1,098
Kerala	2020	Ward member	Open	83.8	7,726
Kerala	2020	Ward member	Women	84.9	8,226
Kerala	2020	District member	Open	86.1	166
Kerala	2020	District member	Women	84.2	165

The confidence intervals reflect regression uncertainty conditional on the extracted and classified data. They do not include transcription error, uncertainty about free-text qualifications, or selection from missing education. Rajasthan’s unspecified qualifications and Kerala’s differential classification rates limit claims about all elected officials. Bihar’s age range is restricted to 21–100, as are the new rural age analyses; the education samples retain those with missing age.

The scripts retain source identifiers, sample counts, recoding dictionaries and typed Parquet analysis files. Input hashes pin the local files used here. Some UP inputs come from integration work whose central adapter has uncommitted changes; the repository preserves the adapter diff and byte hashes rather than presenting that state as a published release. The final data release should reconcile those changes upstream. The included analysis snapshots allow inspection of the fitted samples in the meantime.

The 2021 Bihar candidate release lacks statewide structured winner education, so 2021 enters the age analysis only. Delhi’s linked source profiles cover the recorded winner universe incompletely; the appendix includes every unlinked winner in the missing-record bounds.

The missing-education bounds hold the all-winner regression design fixed and allow each unclassified graduate indicator to be either zero or one. After residualizing reservation on the controls, the coefficient is a weighted sum of outcomes. Assigning missing outcomes according to the signs of those weights gives the exact minimum and maximum coefficient. These bounds concern the adjusted coefficient, not the causal effect, and contain no sampling uncertainty. UP’s bounds remain negative in each election; Rajasthan’s and Kerala’s include both signs.

Table 2: Missing-education bounds on the adjusted all-winner graduate-share difference (percentage points). These are not confidence intervals.

State	Year	Office	Lower	Upper	Missing	N
UP	2010	Village head	-13.5	-12.4	254	51,820
UP	2015	Village head	-9.1	-9.1	0	59,019
UP	2021	Village head	-12.0	-1.3	2,543	49,771
Rajasthan	2020	Village head	-29.7	31.5	3,380	11,128
Kerala	2010	Block member	-12.3	3.4	165	2,078
Kerala	2010	Ward member	-10.6	8.9	1,617	16,641
Kerala	2010	District member	-12.7	4.8	29	331
Kerala	2015	Block member	-13.8	10.1	247	2,062
Kerala	2015	Ward member	-12.2	10.5	1,800	15,921
Kerala	2015	District member	-18.0	3.7	36	331
Kerala	2020	Block member	-14.5	7.5	228	2,078
Kerala	2020	Ward member	-14.4	16.9	2,495	15,952
Kerala	2020	District member	-15.1	15.2	49	331

9 Appendix: complete rural estimates

9.1 Bihar: education

Table 3: Bihar rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2016	Village head	-15.0 [-16.5, -13.4]	1.5 [0.9, 2.1]	7,873	523	Block
2016	Ward member	-4.6 [-4.9, -4.2]	11.5 [10.8, 12.3]	97,200	517	Block
2016	Sarpanch	-12.1 [-13.4, -10.8]	4.7 [3.7, 5.6]	7,841	521	Block
2016	Panch	-2.8 [-3.1, -2.6]	14.8 [14.0, 15.7]	85,355	516	Block
2016	Block member	-10.8 [-12.0, -9.5]	3.2 [2.5, 4.0]	10,743	519	Block
2016	District member	-12.7 [-18.4, -6.9]	1.3 [-0.4, 2.9]	888	38	District

9.2 Uttar Pradesh: education

Table 4: Uttar Pradesh rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Village head	-13.1 [-14.5, -11.7]	6.3 [5.1, 7.4]	51,566	72	District
2015	Village head	-9.1 [-9.7, -8.5]	9.3 [8.7, 10.0]	59,019	823	Block
2021	Village head	-7.4 [-8.1, -6.7]	8.2 [7.5, 9.0]	47,228	728	Block

9.3 Rajasthan: education

Table 5: Rajasthan rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2020	Village head	-6.9 [-9.0, -4.9]	0.1 [-0.1, 0.2]	7,748	352	Block

9.4 Kerala: education

Table 6: Kerala rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included. Illiteracy is unavailable.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Ward member	-0.5 [-2.4, 1.3]	—	15,024	14	District
2010	Block member	-4.1 [-7.9, -0.3]	—	1,913	150	Block
2010	District member	-4.3 [-14.1, 5.6]	—	302	14	District
2015	Ward member	-0.1 [-1.6, 1.3]	—	14,121	14	District
2015	Block member	-1.2 [-5.0, 2.6]	—	1,815	148	Block
2015	District member	-7.8 [-21.1, 5.4]	—	295	14	District
2020	Ward member	2.0 [-0.8, 4.7]	—	13,457	14	District
2020	Block member	-3.5 [-7.5, 0.4]	—	1,850	149	Block
2020	District member	-1.8 [-15.6, 12.1]	—	282	14	District

9.5 Age

Table 7: Age of rural officials: differences in years with pointwise 95% confidence intervals. N: winners with age recorded; G: geographic clusters. Controls match the education models.

State	Year	Office	Difference [95% CI]	N	G
Bihar	2016	Village head	-3.6 [-4.1, -3.2]	7,934	524
Bihar	2016	Ward member	-2.0 [-2.2, -1.8]	97,211	517
Bihar	2016	Sarpanch	-5.6 [-6.2, -5.1]	7,871	521
Bihar	2016	Panch	-2.9 [-3.1, -2.7]	85,376	516
Bihar	2016	Block member	-2.7 [-3.1, -2.3]	10,775	519
Bihar	2016	District member	-4.2 [-5.6, -2.7]	891	38
Bihar	2021	Village head	-3.2 [-3.6, -2.7]	8,050	533
Bihar	2021	Ward member	-0.5 [-0.6, -0.3]	109,501	533
Bihar	2021	Sarpanch	-6.2 [-6.7, -5.7]	8,043	533
Bihar	2021	Panch	-3.2 [-3.4, -3.0]	106,636	533
Bihar	2021	Block member	-2.3 [-2.7, -1.9]	10,972	533
Bihar	2021	District member	-3.9 [-5.0, -2.8]	1,144	38
Uttar Pradesh	2010	Village head	-1.4 [-1.6, -1.2]	51,662	72
Uttar Pradesh	2021	Village head	-1.3 [-1.5, -1.1]	49,757	728
Rajasthan	2020	Village head	-1.7 [-2.3, -1.1]	11,113	352
Kerala	2010	Ward member	-4.8 [-5.9, -3.8]	14,793	14
Kerala	2010	Block member	-4.9 [-5.8, -4.0]	1,980	151
Kerala	2010	District member	-4.2 [-7.3, -1.2]	330	14
Kerala	2015	Ward member	-4.2 [-5.2, -3.2]	15,724	14
Kerala	2015	Block member	-3.9 [-4.9, -3.0]	2,029	151
Kerala	2015	District member	-3.0 [-5.8, -0.3]	327	14
Kerala	2020	Ward member	-3.5 [-4.5, -2.4]	15,860	14
Kerala	2020	Block member	-3.1 [-4.0, -2.2]	2,070	152
Kerala	2020	District member	-3.6 [-6.0, -1.2]	331	14

10 Appendix: Delhi estimates and coverage

Table 8: Delhi councillors: adjusted differences in percentage points, except age in years. Assembly-constituency and caste-reservation controls; pointwise intervals clustered by assembly constituency. G denotes clusters.

Year	Outcome	Difference [95% CI]	N	G
2012	Graduate or above	-0.6 [-14.4, 13.2]	226	65
2012	Illiterate	2.5 [-1.3, 6.3]	226	65
2012	Age (years)	-6.0 [-9.0, -3.0]	235	65
2012	Any pending case	-23.5 [-35.2, -11.8]	237	65
2017	Graduate or above	1.1 [-10.4, 12.6]	262	68
2017	Illiterate	0.6 [-1.7, 2.8]	262	68
2017	Age (years)	-3.6 [-5.6, -1.5]	262	68
2017	Any pending case	-10.6 [-18.4, -2.8]	267	68
2022	Graduate or above	-3.4 [-15.9, 9.1]	244	68
2022	Illiterate	2.4 [-1.0, 5.7]	244	68
2022	Age (years)	-2.3 [-4.8, 0.3]	248	68
2022	Any pending case	-21.0 [-29.9, -12.0]	248	68

Table 9: Delhi outcome coverage among recorded winners. Percentages use all winners of each seat type as the denominator.

Year	Seat	N	Education (%)	Age (%)	Cases (%)
2012	Open	134	85.8	86.6	87.3
2012	Women	138	80.4	86.2	87.0
2017	Open	134	97.0	98.5	99.3
2017	Women	138	95.7	94.2	97.1
2022	Open	125	97.6	100.0	100.0
2022	Women	125	97.6	98.4	98.4

Table 10: Delhi missing-outcome bounds on the adjusted all-winner coefficient, in percentage points. These are not confidence intervals and allow no sampling or transcription uncertainty.

Year	Outcome	Lower	Upper	Missing	N
2012	Graduate or above	-16.4	18.9	46	272
2012	Any pending case	-33.9	-6.1	35	272
2017	Graduate or above	-2.5	6.3	10	272
2017	Any pending case	-11.5	-7.5	5	272
2022	Graduate or above	-5.0	-0.1	6	250
2022	Any pending case	-21.3	-19.9	2	250

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